OBSERVABLE CLOUD SERVICE

Instrumented & Monitored Cloud Service

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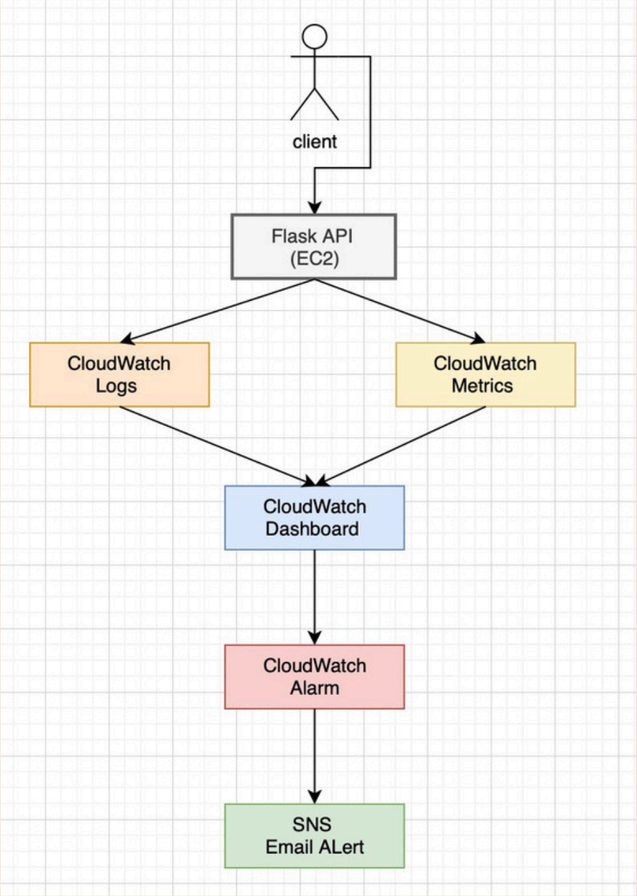
Course: Cloud Engineering Bootcamp - Week 3
Date: March 13, 2026

Goal of the Project

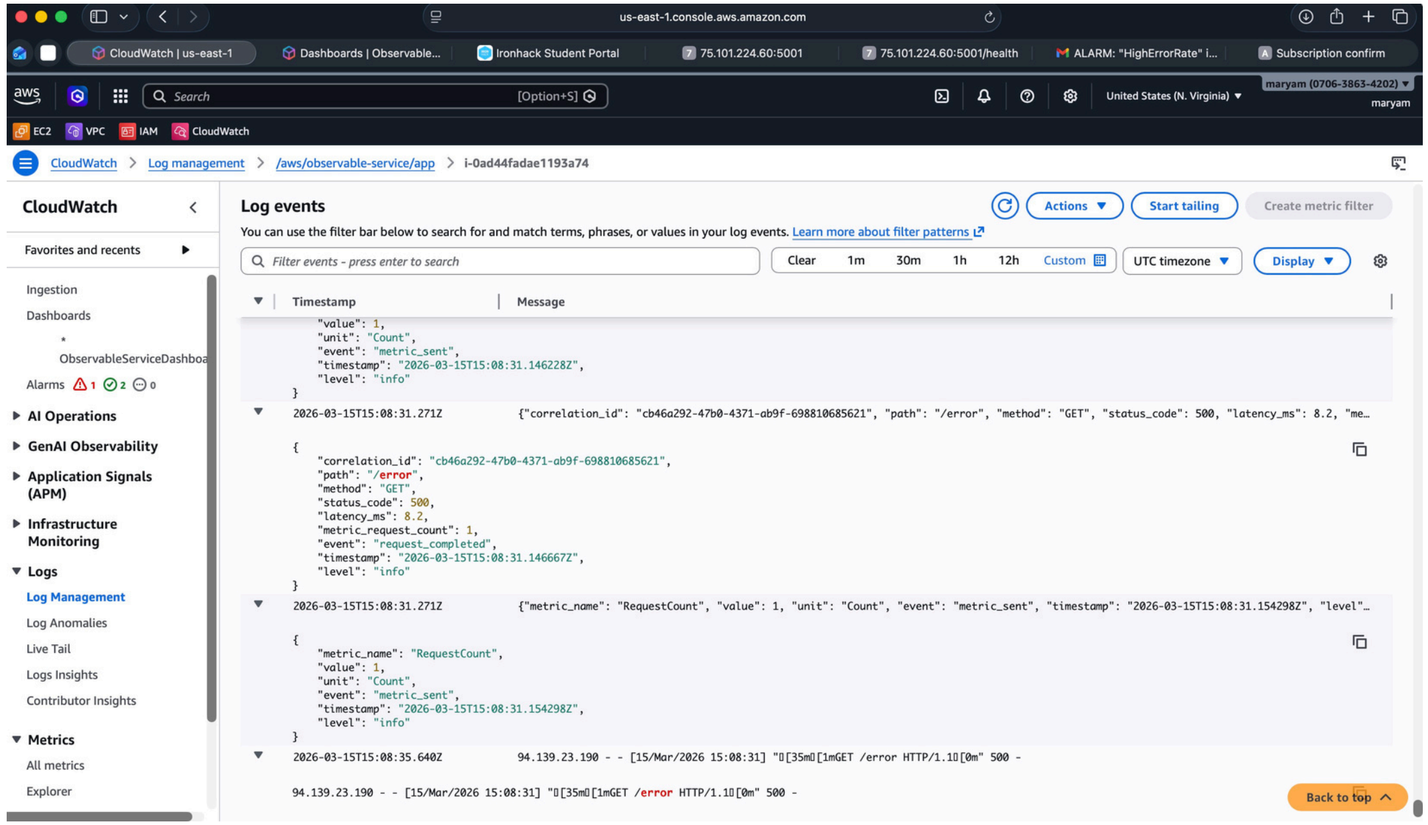
- Build a production-style observable system
- Implement structured logging
- Collect custom application metrics
- Monitor system health using dashboards
- Detect failures using automated alerts

System Architecture



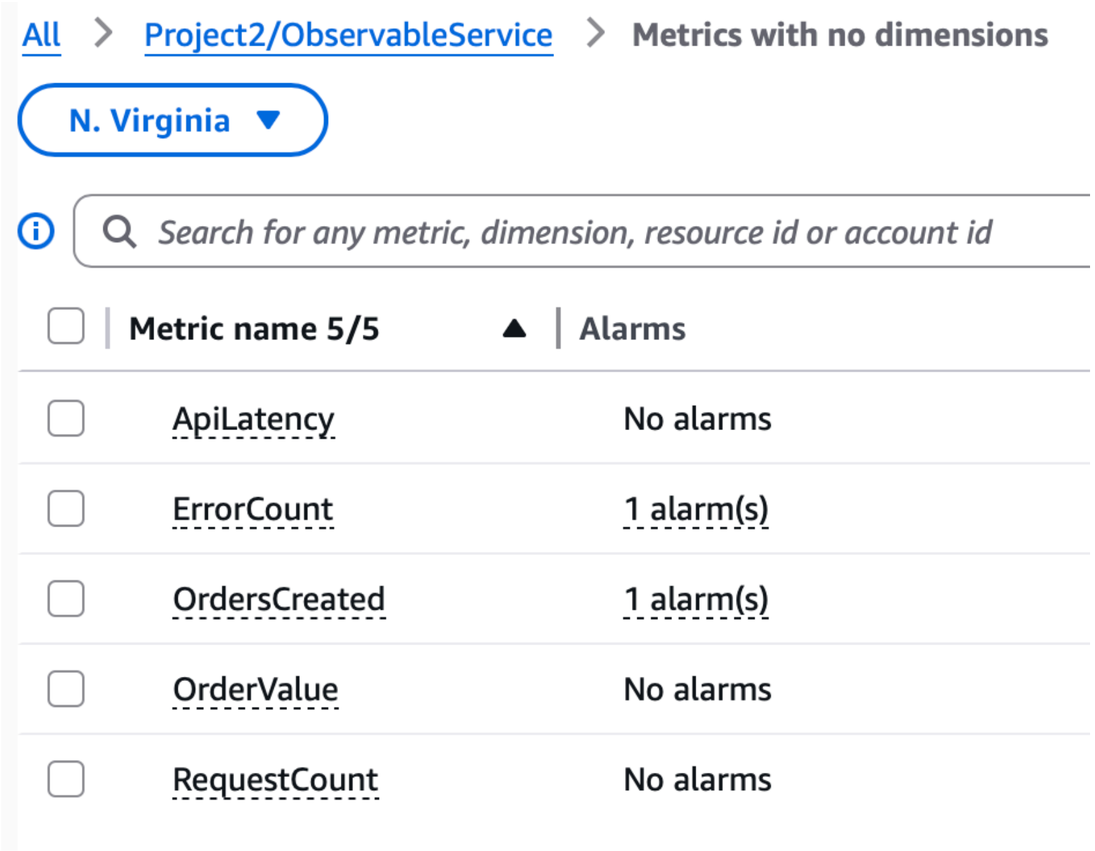
- Flask API running on EC2
- CloudWatch Logs for centralized logging
- CloudWatch Metrics for application telemetry
- CloudWatch Dashboard for visualization
- SNS for alert notifications

Structured Logging



- JSON structured logs using structlog
- Each request assigned a correlation_id

Custom Metrics



Custom metrics provide real-time insight into application behavior

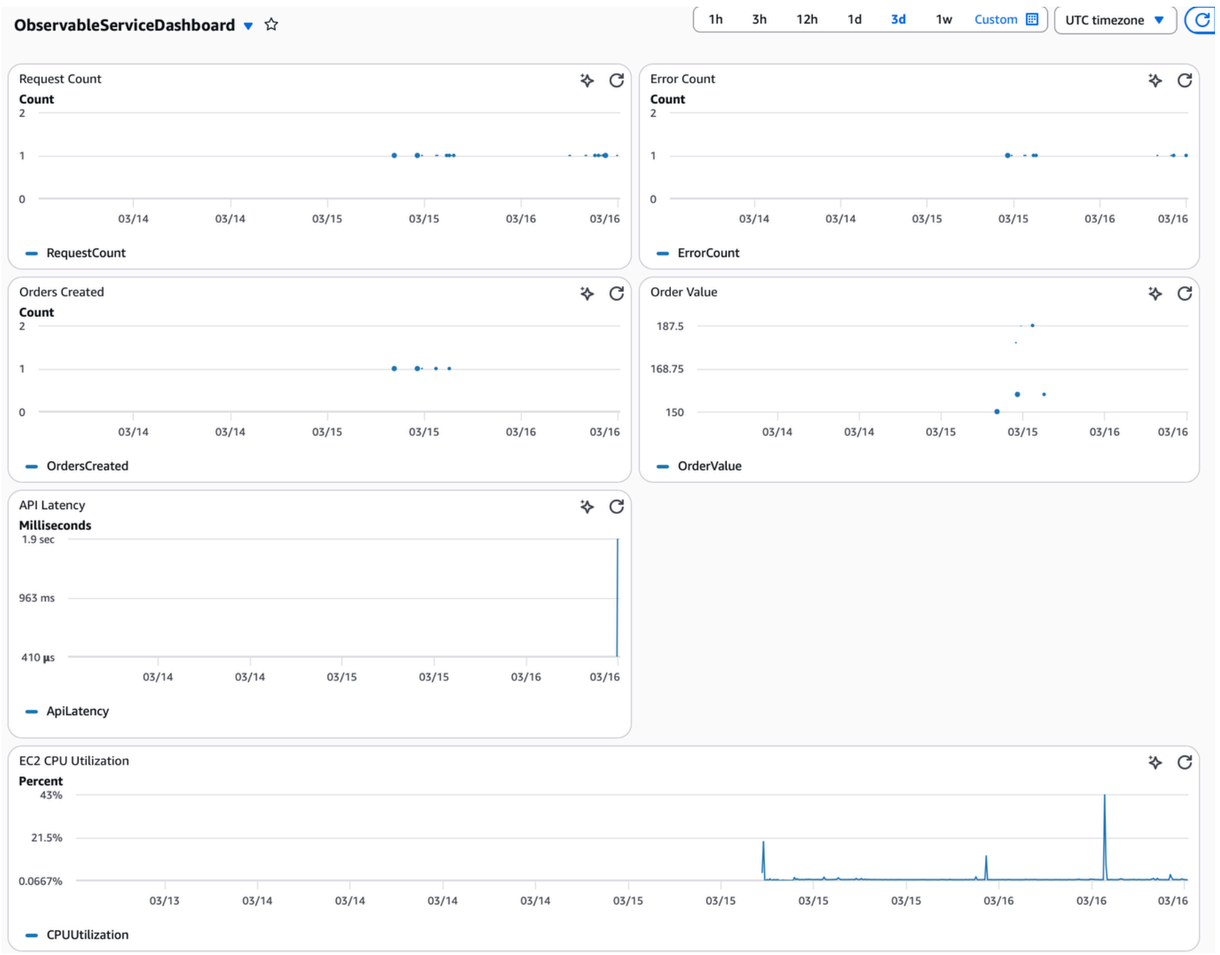
Monitoring Dashboard

Golden Signals:

- Request Rate → Traffic
- Error Rate → Errors
- Latency
- Saturation

Widgets included:

- RequestCount
- ErrorCount
- OrdersCreated
- OrderValue
- API latency
- CPUUtilization



Alerting System

CloudWatch > Alarms

CloudWatch console feature

Alarms | Mute rules - *new*

Alarms (3)

Clear selection | Create composite alarm | Actions | Create alarm

Filter alarms | Quick filters

Name	State	Actions	Actions muted - *new*	Last state update (UTC)	Conditions
HighErrorRate	In alarm	Actions enabled	-	2026-03-16 10:19:08	ErrorCount > 5 for 1 datapoints within 1 minute
LowOrders	Insufficient data	Actions enabled	-	2026-03-15 15:23:58	OrdersCreated < 1 for 1 datapoints within 5 minutes
HighCPU	OK	Actions enabled	-	2026-03-15 13:19:54	CPUUtilization > 70 for 1 datapoints within 5 minutes

Alarm	Condition
HighCPU	CPU > 70%
HighErrorRate	ErrorCount > 5
LowOrders	OrdersCreated < 1

ALARM: "HighErrorRate" in US East (N. Virginia)

4 messages

observable-service-alerts <no-reply@sns.amazonaws.com> Mon, Mar 16, 2026 at

To: maryamahmadi.apply@gmail.com

You are receiving this email because your Amazon CloudWatch Alarm "HighErrorRate" in the US East (N. Virginia) region has entered the ALARM state, because "Threshold Crossed: 1 out of the last 1 datapoints [36.0 (16/03/26 07:55:00)] was greater than the threshold (5.0) (minimum 1 datapoint for OK -> ALARM transition)." at "Monday 16 March, 2026 07:56:08 UTC".

View this alarm in the AWS Management Console:
https://us-east-1.console.aws.amazon.com/cloudwatch/deeplink.js?region=us-east-1#alarmsV2:alarm/HighErrorRate

Alarm Details:
- Name: HighErrorRate
- Description: Alarm when error rate exceeds 5
- State Change: INSUFFICIENT_DATA -> ALARM
- Reason for State Change: Threshold Crossed: 1 out of the last 1 datapoints [36.0 (16/03/26 07:55:00)] was greater than the threshold (5.0) (minimum 1 datapoint for OK -> ALARM transition).
- Timestamp: Monday 16 March, 2026 07:56:08 UTC
- AWS Account: 070638634202
- Alarm Arn: arn:aws:cloudwatch:us-east-1:070638634202:alarm:HighErrorRate

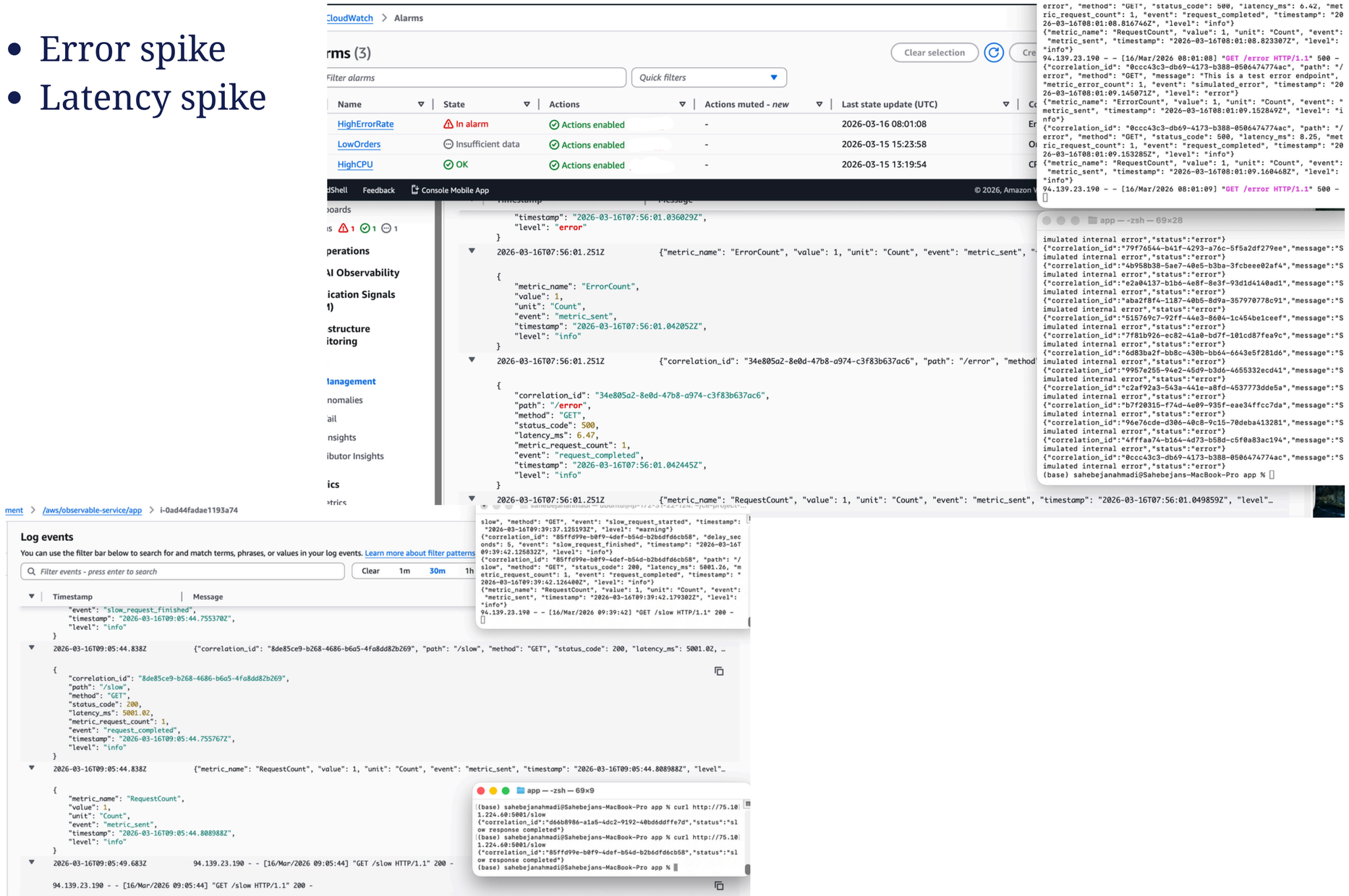
Threshold:
- The alarm is in the ALARM state when the metric is GreaterThanThreshold 5.0 for at least 1 of the last 1 period(s) of 60 seconds.

Monitored Metric:
- MetricNamespace: Project2/ObservableService
- MetricName: ErrorCount
- Dimensions:
- Period: 60 seconds
- Statistic: Sum
- Unit: not specified
- TreatMissingData: missing

State Change Actions:
- OK:
- ALARM: [arn:aws:sns:us-east-1:070638634202:observable-service-alerts]
- INSUFFICIENT_DATA:

Failure Simulation & Incident Investigation

- Error spike
- Latency spike



Root Cause Analysis


```
{"correlation_id": "e2ff116d-8423-4205-b2ba-8fd1c04e6401", "path": "/error", "method": "GET", "message": "This is a test err
or endpoint", "metric_error_count": 1, "event": "simulated_error", "timestamp": "2026-03-16T09:56:12.905467Z", "level": "err
or"}
{"metric_name": "ErrorCount", "value": 1, "unit": "Count", "event": "metric_sent", "timestamp": "2026-03-16T09:56:12.971143Z
", "level": "info"}
{"correlation_id": "e2ff116d-8423-4205-b2ba-8fd1c04e6401", "path": "/error", "method": "GET", "status_code": 500, "latency_m
s": 66.5, "metric_request_count": 1, "event": "request_completed", "timestamp": "2026-03-16T09:56:12.971728Z", "level": "inf
o"}
{"metric_name": "RequestCount", "value": 1, "unit": "Count", "event": "metric_sent", "timestamp": "2026-03-16T09:56:12.97789
9Z", "level": "info"}
94.139.23.190 - - [16/Mar/2026 09:56:12] "GET /error HTTP/1.1" 500 -
```


- Investigated CloudWatch Logs
- Identified repeated requests to /error
- Application returned HTTP 500 responses
- ErrorCount metric increased
- Root cause traced to simulated error endpoint
- Correlation ID used to trace the request lifecycle

Q&A

THANK YOU